Question-1→ Explain the all selectors used in CSS?

Solution-1→ CSS (Cascading style sheets) selectors are patterns used to select the elements you want to style on a web page. Understanding the different types of selectors is crucial for efficient and maintainable CSS.

Here in CSS mainly five selectors are :-

1- Simple Selectors
2- Combinator selectors
3- Attribute selectors
4- Pseudo - class selectors
5- Pseudo - element selectors

1- Simple selectors :- Simple selector are the most basic and common type of selectors. They target a single HTML element based on its name, class or ID.

a- Type selector (Element selector):- This selector targets all elements of a specific HTML type.

Syntax :- element_name

Example-
```
p {
    font-family : sans-serif;
    Colour : #333;
}
```

b- class selector :- This selector targets all elements with a specific class attribute. Classes are great for applying styles to multiple, non-consecutive element.

Syntax- .class_name

Example-
```
<h2 class = "section-title" > Latest News </h2>
<p class = "highlight" > This is an important point </p>
```

```
.section-title {
    color: navy;
    border-bottom: 2px solid blue;
}
.highlight {
    background-color: yellow;
}
```

c- **ID selector**:- This selector targets a single, unique element with a specific id attribute. An id should be unique within a page.

Syntax- #id_name

Example- **Html code**

```
<div id="main-header">
    <h1> My Website </h1>
</div>
```

css code

```
# main-header {
    background-color: #f0f0f0;
    padding: 20px;
}
```

d- **Universal selector** - This selector targets every single element on the page. It's often used for resetting default browser styles.

Syntax- *

Example-

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}
```

2- **Combinator selector**:- Combinator selectors are used to select elements based on their relationship or position relative to other elements in the HTML structure.

a- **Descendant selector:-** This selector targets all the elements that are descendants (children, grandchildren, etc) of a specific element.

Syntax - ansestor_selector descendant_selector (separated by a space)

Example - html code

```
<div class="card">
  <h2> Card Title </h2>
  <p> Card content </p>
  <a href="#"> Read more </a>
</div>
```

css code

```
.card p {
  line-height: 1.6;
  font-size: 14px;
}
```

b- **Child selector** - This selector targets elements that are direct children of a specified parent element.

Syntax - parent_selector > child_selector

Example - html code

```
<ul>
  <li> item 1 </li>
  <li> item 2 </li>
  <ul>
    <li> sub-item </li>
  </ul>
</ul>
```

css code

```
ul > li {
  list-style-type: none;
}
```

c- **Adjacent sibling selector** - This selector targets an element that is immediately preceded by a specific sibling element.

Syntax :- element_1 + element_2

Example - html code

```
<p> This a Paragraph </p>
<h2> This is a heading </h2>
<p> This paragraph comes again </p>
```

css code

```
h2 + p {
  margin-top: 0;
  font-style: italic;
}
```

d-General sibling selector - This selector targets all elements that are siblings of a specified element regardless of their position.

Syntax:- element_1 ~ element_2

Example - html code

```
<h1> My heading </h1>
<p> First paragraph </p>
<h2> Sub-heading </h2>
<p> Second Paragraph </p>
```

css code

```
h1 ~ p {
  color: grey;
}
```

3-Albribute selector - Attribute selector in CSS allow you to target and style HTML elements based on the presence or value of their attributes.

- [attribute]:- selects any element that has specified attribute, regardless of its values.
- [attribute = "value"]:- selects elements where the attributes has an exact, specific value.
- [attribute ^= "value"]:- selects elements where the attributes value begins with a specific string.

Example- html code

```
<a href="https://example.com/about">About</a>
<a href="/contact">Contact</a>
```

Css code

```
a[href ^= "https"] {
  color: green;
}
a[href $= "Contact"] {
  background-color: lightblue;
```

4- **Pseudo-class selectors** - Pseudo-class selector are used to select elements based on a state or position, rather than a fixed structure. They begin with a colon (:).

a- **State Pseudo-classes (User Action)**

These select elements based on user interaction.

- :hover → When the mouse is over an element
- :active → when an element is clicked.
- :focus → when an element (like an input field) has focus.
- :visited → for links that have been visited.
- :link → for unvisited links

Example→

```
a:hover {
    color: red;
    text-decoration: none;
}
```

```
input:focus {
    border-color: blue;
    box-shadow: 0 0 5px blue;
}
```

b- **structural Pseudo-classes (Position)**:- These selects element based on their position within a parent element

- :first-child → the first child of its parent.
- :last-child → the last child of its parent.
- :nth-child(n) → The nth child of its parent
- :first-of-type → the first element of its type within its parent.
- :last-of-type → the last element of its type within its parent.
- :n^{th}-of-type(n) → the nth element of its type within its parent.

Example- html code

```
<ul>
    <li> item 1 </li>
    <li> item 2 </li>
    <li> item 3 </li>
</ul>
```

css code

```
li:first-child {
    font-weight: bold;
}
li:nth-child(even) {
    background-color: #eee;
}
```

5- Pseudo-Elements Selectors- Pseudo-elements are used to style a specific part of an element. They are represented by two colons (::)

- ::before → inserts content before the content of an element.
- ::after → inserts content after the content of an element
- ::first-line → styles the first line of a block-level element
- ::first-letter → styles the first letter of block-level element
- ::selection → styles the text selected by the user.

Example:- html code

```
<p> This is a paragraph </p>
```

css code

```
p::first-letter {
  font-size: 3rem;
  float: left;
  line-height: 1;
  margin-right: 5px;
}
```

Question-2- Explain all protocols in web pages?

Solution- Web pages rely on a suite of protocols to function working, together to get content from a server to your browser. Here are some protocols involved:-

(i)- HTTP (Hypertext Transfer Protocol):- This is the foundation of World Wide Web. It's a client-server protocol used to request and receive web pages and resources. When you type a URL into your browser, it sends an HTTP request to web server and the server sends back an HTTP response with webpage content.

(ii)- HTTPS (Hypertext Transfer Protocols Secure):- This is the secure version of the

HTTP. It uses SSL/TLS (Secure Sockets layer/ Transport layer security) to encrypt the data exchanged between your browser and the web server. You can identify HTTPS by the "https://" at the beginning of the URL and a padlock icon in your browser address bar.

(iii)- TCP/IP (Transmission Control Protocol / Internet Protocol):-
This is a core protocol suit of the internet.

- IP (Internet Protocol)- Responsible for addressing and routing data packets across the network. It's like the postal system for the internet, ensuring that data packets get to correct destination using IP addresses.
- TCP (Transmission Control Protocol):- Works on top of IP to ensure reliable, ordered and error-checked delivery of data. It breaks down large pieces of data into smaller packets, sends them and then resassembles them in correct order at destination. It also handles re-sending any packets that get lost along the way.

(iv)- DNS (Domain Name System):- The internet's "phonebook". DNS translates human-readable domain names (like www.google.com) into numerical IP addresses that computers use to find each other on network. When you enter a URL, your computer first contacts a DNS server to get the IP addresses of the website you want to visit.